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Midnight Chefs

LEFT OVER FOOD RECIPE

A refreshing and tangy twist for your meal

Problem Statement



- Random leftover ingredients are often wasted
- Edible food gets thrown away
- Lack of recipe ideas
- Navigating complex dietary restrictions and food allergies, leading to unnecessary food waste, meal planning friction, and potential health risks.





Solution



SMART RECIPE GENERATOR

- Uses **Vision AI** to instantly scan **leftover ingredients**, filters them against **user-specific allergy** profiles
- Automatically suggests **safe, nutritionally balanced recipe alternatives** with smart ingredient substitutions.
- Provides **quick and simple recipes**
- **Nutrition chart**
- Save recipes

FEATURES

Input

- Ingredients Image scanner
- Recipe Category
- Cooking Time

Nutrition chart

- Calories
- Protein
- Carbohydrates
- Fat , Fiber
- Sugar

Allergens

- User can input the allergens
- Smart substitution for allergens

Why This is Special

- Dedicated short snippet explaining the specific flavor harmony of the combination.

TECHNOLOGY

Vision AI Engine

- **Frameworks:** OpenCV, YOLOv8 / PyTorch
- **Function:** Real-time ingredient recognition & object detection from user camera feeds
- **Processing:** Edge inference + Cloud fallback

Allergy & Nutrition Engine

- **Logic Core:** Python & Rule-based expert systems
- **Function:** Cross-references allergens with macro-nutrient substitution databases
- **Database:** USDA Food Data Central API

Backend & API Layer

- **Runtime:** Node.js, Express [FastAPI]
- **Database:** PostgreSQL & Redis (Caching recipe data)
- **Cloud:** AWS / GCP Serverless Functions



Approach



1. **Input Capture & User Profiling:**

- a. Users capture a photo of their leftover ingredients using camera.
- b. inputting their personal food preferences, and allergy profiles.

2. **Vision AI Recognition:**

- a. Computer vision models (OpenCV and YOLOv8) process the camera stream to instantly detect, segment, and tag all visible food items with high accuracy.

3. **Allergy Cross-Referencing:**

- a. Title & Description
- b. Precise Ingredient Breakdown
- c. Speciality description



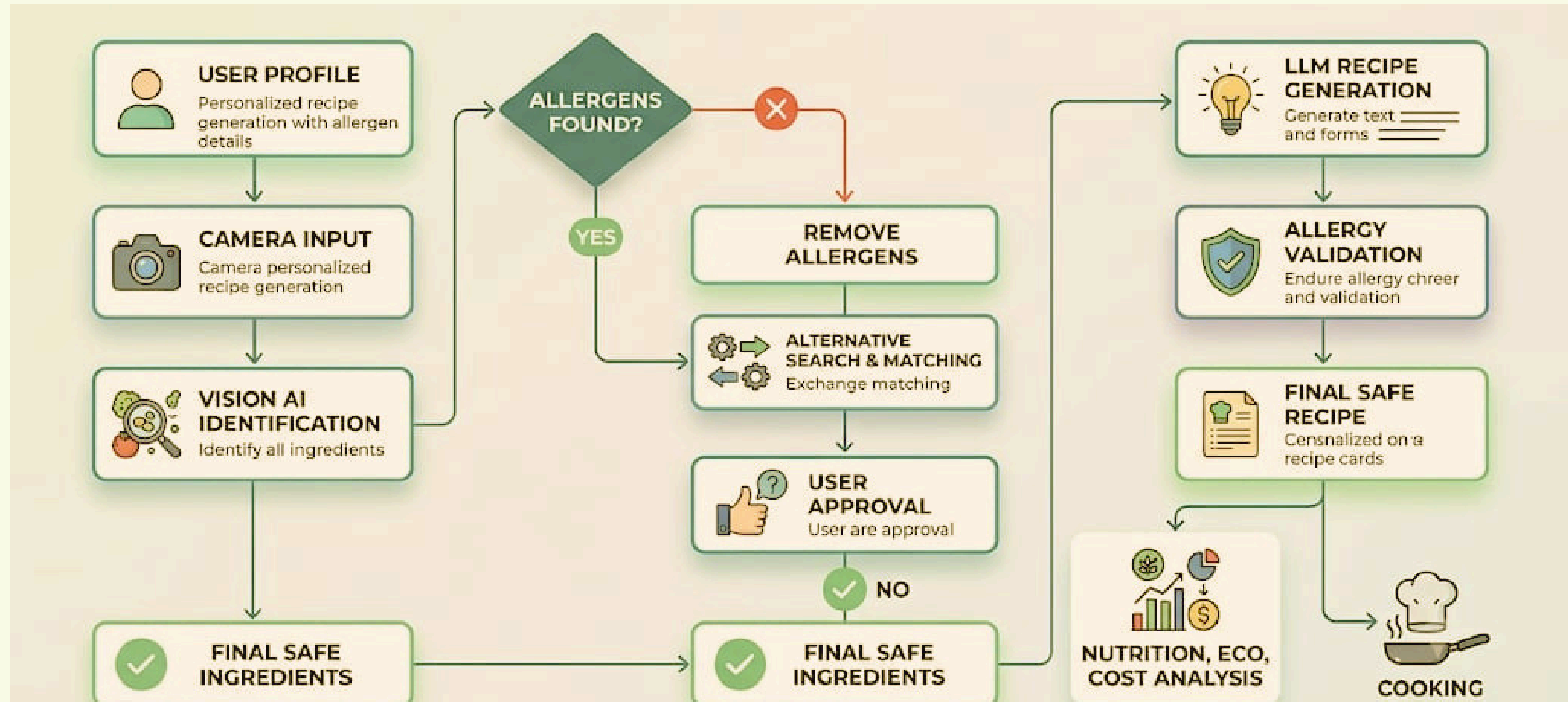
4. Nutritional Substitution Engine:

- a. Using **USDA nutritional databases**, the system **automatically replaces** restricted or missing **ingredients with safe**, macro-nutritionally balanced alternatives.

5. Recipe Generation & Output:

- a. The app compiles the verified ingredients into customized, step-by-step cooking instructions optimized to minimize household food waste.

FLOWCHART



DATASET

MULTI-SOURCE INGREDIENT INTELLIGENCE

250K+

MAPPED INGREDIENT NODES

Our system integrates INDORI, Food.com, and USDA FDC datasets to provide global culinary coverage and precise nutritional mapping for surplus recovery.



KEY INTEGRATION PILLARS

Culinary Versatility
Nutritional Mapping

**High
Precise**



Midnigh Chefs

THANK YOU!

Stay healthy and keep cooking !

